

修士論文

Microlocal Morse Theory and Truncated Microsupport

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2024.3

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数学専攻博士前期課程

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1 Preface

In this thesis, We study microlocal Morse theory, which is presented, for instance, in [GKS12]. However, we replace the notion of microsupports with that of truncated microsupprts, introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KFS03]. In [KFS03], they do not study the functorial property of truncated microsupport of direct images of non proper morphisms. We prove this functorial property under non proper direct images.

2 Notations and Reviews

We consider a commutative unital ring $\mathbf{k}$ of finite global dimension (e.g. $\mathbf{Z}$ or a field). Let X be a real analytic manifold. We denote the cotangent bundle of X by $\pi: T^*X \rightarrow X$. We denote by $D^b(\mathbf{k}_X)$ the bounded derived category of sheaves of $\mathbf{k}$ -modules on X .

Let us denote Z a locally closed set of X and $i: Z \hookrightarrow X$ an inclusion. We use the following functors:

$$R\Gamma_Z(\cdot) := Ri_*i^!, \quad (\cdot)_Z := Ri_!i^{-1}.$$

Geometrical Notations

Let M be a manifold. We denote by $\pi: T^*M \rightarrow M$ the cotangent bundle of M . If $f: M \rightarrow N$ is a morphism of manifolds, then we have the natural diagram.

$$\begin{array}{ccccc} TM & \xrightarrow{f'} & M \times_N TN & \xrightarrow{f_\tau} & TN \\ & \searrow \tau_M & \downarrow N & \square & \downarrow \tau_N \\ & & M & \xrightarrow{f} & N, \end{array} \quad \begin{array}{ccccc} T^*M & \xleftarrow{f_d} & M \times_N T^*N & \xrightarrow{f_\pi} & T^*N \\ & \searrow \pi_M & \downarrow N & \square & \downarrow \pi_N \\ & & M & \xrightarrow{f} & N. \end{array}$$

In this diagram, f_τ and f_π denote the natural projections. f' is defined by $(x; v) \mapsto (x; df_x v)$, and f_d is induced by the transposition of f' .

3 Truncated Microsupport

Definition 3.1. Let $F \in D^b(\mathbf{k}_X)$. The closed conic subset $SS_k(F)$ of T^*X is defined by: $p \notin SS_k(F)$ if and only if there exists an open conic neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1 -function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$(3.1) \quad H_{\{\varphi \geq 0\}}^j(F)_x = 0 \quad \text{for any } j \leq k.$$

3.1 Functorial Properties

Proposition 3.2. Let $f: X \rightarrow Y$ be a morphism of real analytic manifolds and let $F \in D^b(\mathbf{k}_X)$ such that f is proper on $\text{supp}(F)$. Then for any $k \in \mathbf{Z}$,

$$(3.2) \quad SS_k(Rf_* F) \subset f_\pi f_d^{-1}(SS_k(F)).$$

The equality holds in case f is a closed embedding.

4 Lemmas from [KS90]

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary.

For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$(4.1) \quad E_n^\bullet: \cdots \rightarrow M_n^{j-1} \rightarrow M_n^j \rightarrow M_n^{j+1} \rightarrow \cdots,$$

and projective limit

$$(4.2) \quad E_\infty^\bullet: \dots \rightarrow M_\infty^{j-1} \rightarrow M_\infty^j \rightarrow M_\infty^{j+1} \rightarrow \dots$$

Denote by

$$(4.3) \quad \Phi_k: H^k(E_\infty^\bullet) \rightarrow \varprojlim_n H^k(E_n^\bullet)$$

the natural morphism.

Proposition 4.1 ([KS90, Proposition 1.12.4]). Assume that for each $j \in \mathbf{Z}$, the system $(M_n^j)_n$ satisfies the ML condition. Then

- (a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,
- (b) if moreover, for a given i the system $(H^{i-1}(E_n^\bullet))_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 4.2 ([KS90, Proposition 1.12.6]). Let $(X_s, \rho_{s,t}: X_t \rightarrow X_s)_{(s,t) \in \mathbf{R}, (s \leq t)}$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s: X_s \rightarrow \varprojlim_{r < s} X_r$$

and

$$\mu_s: \varinjlim_{t > s} X_t \rightarrow X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0, s_1} ($s_0 \leq s_1$) are injective (resp. surjective).

ML procedure for sheaves

Proposition 4.3 ([KS90, Proposition 2.7.1]). Let X be a topological space and let $F \in \mathbf{D}^+(\mathbf{Z}_X)$. Let $(U_n)_{n \in \mathbf{N}}$ be an increasing sequence of open subsets of X , $(Z_n)_{n \in \mathbf{N}}$ a decreasing sequence of closed subsets of X . Set $U := \bigcup_n U_n$, $Z := \bigcap_n Z_n$.

- (i) For any j , the natural map $\varphi_j: H_Z^j(U; F) \rightarrow \varprojlim_n H_{Z_n}^j(U_n; F)$ is surjective.
- (ii) Assume that for a given j , the projective system $(H_{Z_n}^{j-1}(U_n; F))_n$ satisfies the ML condition. Then φ_j is bijective.
- (iii) Let now $(X_n)_{n \in \mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \text{Int}(X_{n+1})$ for all n . Assume that for a given j , the projective system $(H^{j-1}(X_n; F))_n$ satisfies the ML condition. Then the natural map $H^j(X; F) \rightarrow \varprojlim_n H^j(X_n; F)$ is bijective.

5 Results

In this section, let M be a manifold.*¹

*¹Can M be of class C^∞ ?

Theorem 5.1. Let F in $D^b(\mathbf{k}_M)$, and $\psi: M \rightarrow \mathbf{R}$ be a real-valued function of class C^1 , proper on $\text{supp}(F)$. Let $a < b$ be real numbers. We assume $d\psi(x) \notin \text{SS}_k(F)$ for $a \leq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t := \psi^{-1}(]-\infty, t[)$. Then, the restriction morphisms

$$H^j(M_b; F) \rightarrow H^j(M_a; F)$$

are isomorphic for each $j < k$.^{*2}

Lemma 5.2. Let $-\infty < a < b < +\infty$ be real numbers and $G \in D^b(\mathbf{R})$. Let k be an integer. If

$$\left(H^j_{[t, +\infty[}(G) \right)_t \cong 0 \quad \text{for all } t \in [a, b[, \text{ and all } j \leq k.$$

Then one has

$$H^j(]-\infty, b[; G) \xrightarrow{\sim} H^j(]-\infty, a[; G)$$

for all $j < k$.^{*3}

Proof. Let $t \in [a, b[$. We will denote by I_t the open interval $]-\infty, t[$ and denote by Z_t the closed interval $]-\infty, t[(= \overline{I_t})$. By applying the functor $(\cdot)_{\overline{I_t}}$ to the distinguished triangle

$$\text{R}\Gamma_{\mathbf{R}-I_t}(G) \rightarrow G \rightarrow \text{R}\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } D^b(\mathbf{k}_{\mathbf{R}}),$$

we have

$$(\text{R}\Gamma_{[t, +\infty[}(G))_{\{t\}} \rightarrow G_{Z_t} \rightarrow \text{R}\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } D^b(\mathbf{k}_{\mathbf{R}}).$$

Applying the functor $\text{R}\Gamma(\mathbf{R}; \cdot)$ to this, we have

$$(5.1) \quad (\text{R}\Gamma_{[t, +\infty[}(G))_t \rightarrow \text{R}\Gamma(Z_t; G) \rightarrow \text{R}\Gamma(I_t; G) \xrightarrow{+1} \quad \text{in } D^b(\mathbf{k}).$$

Then consider the following conditions for $j \in \mathbf{Z}$, $s \in [a, b[$, and $t \in]a, b[$:

$$\begin{aligned} (a)_s^j & \quad \varinjlim_{t>s} H^j(I_t; G) \xrightarrow{\sim} H^j(I_s; G), \\ (b)_t^j & \quad \varprojlim_{s<t} H^j(I_t; G) \xleftarrow{\sim} H^j(I_t; G). \end{aligned}$$

By hypothesis, it follows from (5.1) that $H^j(Z_t; G) \xrightarrow{\sim} H^j(I_t; G)$ for $t \in [a, b[$.^{*4} Since $\varinjlim_{t>s} H^j(I_t; G) \cong H^j(Z_s; G)$, (a)_s^j holds for any $j < k$ and $s \in [a, b[$.^{*5} Moreover, since we assume G is a bounded complex,

^{*2}For $j = k$, this morphism might be injective.

^{*3}For k , this morphism might be injective. (not written in this draft.)

^{*4}Similarly, we have injections $H^k(Z_t; G) \hookrightarrow H^k(I_t; G)$ for $t \in [a, b[$.

^{*5}Similarly,

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(Z_s; G)$$

gives injections

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(I_s; G)$$

for all $s \in [a, b[$.

$(b)_t^j$ holds for $j \ll 0$ and for all $t \in]a, b]$. Let us denote such j by j_0 . Let us argue by induction on j . Assume $(b)_t^j$ holds for all $t \in]a, b[$ and all $j \leq j_0$. Applying Proposition 4.2, we find the isomorphisms

$$(5.2) \quad H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j \leq j_0 \text{ and all } a < s \leq t < b.$$

Let $G \xrightarrow{\sim} G^\bullet$ be a flabby resolution. Let $t \in]a, b]$. Consider a complex

$$E_n^\bullet = \Gamma(I_{t-\frac{1}{n}}; G^\bullet).$$

Since $G^\bullet$ is flabby, we have $E_n^j \cong E_\infty^j$ and the projective system $(E_n^j)_n$ satisfies the ML condition for all $j \in \mathbf{Z}$.

By (5.2), the projective system $(H^{j_0}(E_n^\bullet))_n$ satisfies the ML condition (because we have $a < t - \frac{1}{n} < t$ for $n \gg 0$). Applying Proposition 4.1 to $(E_n^\bullet)_n$, it follows that $(b)_t^j$ holds for $j_0 + 1$ and the induction proceeds. More precisely, we have

$$\begin{aligned} H^{j_0+1}(I_t; G^\bullet) &\xrightarrow{\sim} \varprojlim_{\Phi_{j_0+1}} H^{j_0+1}(I_{t-\frac{1}{n}}; G^\bullet) \\ &\cong \varprojlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G) \\ &\cong \varprojlim_{s < t} H^{j_0+1}(I_s; G). \end{aligned}$$

Again by Proposition 4.2, we get the isomorphisms

$$H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j < k \text{ and all } a < s \leq t \leq b,$$

and injections

$$H^j(I_s; G) \hookrightarrow H^j(I_t; G)$$

for $j = k$ and for all $a < s \leq t \leq b$.

Finally, using (5.1) and the hypothesis, we have the isomorphisms

$$H^j(I_a; G) \xleftarrow{\sim} H^j(Z_a; G) \xrightarrow{\sim} H^j(I_s; G) \quad \text{for all } j < k \text{ and all } a < s \leq b.$$

Similarly, for $j = k$, we have

$$H^k(I_b; G) \cong \varinjlim_{t > s} H^k(I_t; G) \hookrightarrow H^k(I_s; G) \hookrightarrow H^k(I_a; G).$$

□

Proof of Theorem 5.1. First, we have

$$H^j(M_t; F) \cong H^j \mathrm{R}\Gamma([- \infty, t[; \mathrm{R}\psi_* F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leq \psi(x) < b$, and by (3.2), the formula $\mathrm{SS}_k(\mathrm{R}\psi_* F) \subset \psi_\pi \psi_d^{-1} \mathrm{SS}_k(F)$ holds. Hence we have

$$(H^j \mathrm{R}\Gamma_{[\psi(x), +\infty[}(\mathrm{R}\psi_* F))_{\psi(x)} = 0$$

for each $j \leq k$. Then we can apply Lemma 5.2 and have

$$H^j([- \infty, b[; \mathrm{R}\psi_* F) \xrightarrow{\sim} H^j([- \infty, a[; \mathrm{R}\psi_* F)$$

for all $j < k$. Then we can conclude that

$$H^j(M_b; F) \xrightarrow{\sim} H^j(M_a; F)$$

for all $j < k$. □

6 Truncated Noncharacteristic Deformation

We will reformulate the noncharacteristic deformation lemma [KS90, Proposition 2.7.2] to make a relation to truncated microsupport.

Proposition 6.1. Let X be a Hausdorff space, $F \in \mathbf{D}^b(\mathbf{Z}_X)$, and let $(U_t)_{t \in \mathbf{R}}$ be a family of open subsets of X . We assume the following conditions.

- (i) $U_t = \bigcup_{s < t} U_s$, for all $t \in \mathbf{R}$.
- (ii) For all pairs (s, t) with $s \leq t$, the set $\overline{U_t - U_s} \cap \text{supp } F$ is compact.
- (iii) Setting $Z_s := \bigcap_{t > s} \overline{U_t - U_s}$, we have for all pairs (s, t) with $s \leq t$, and all $x \in Z_s - U_t$,

$$(H_{X - U_t}^j(F))_x = 0 \quad \text{for all } j \leq k.$$

- (iv) For any $s \in \mathbf{R}$, $\bigcap_{t > s} U_t = \overline{U_s}$ and $\overline{U_s}$ is closed in a paracompact open subset of X .
- (v) ^{*6} $Z_s - U_s \subset \partial U_s$ for all $s \in \mathbf{R}$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j \left(\bigcup_s U_s; F \right) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for all } j < k,$$

and the injections

$$H^k \left(\bigcup_s U_s; F \right) \hookrightarrow H^k(U_t; F).$$

Proof. Consider the following conditions for $j \in \mathbf{Z}$, s , and t :

$$\begin{aligned} (a)_s^j & \quad \mu_s^j: \varinjlim_{t > s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F), \\ (a')_s^k & \quad \mu_s^k: \varinjlim_{t > s} H^k(U_t; F) \hookrightarrow H^k(U_s; F), \\ (b)_s^j & \quad \lambda_s^j: H^j(U_t; F) \xrightarrow{\sim} \varprojlim_{s < t} H^j(U_s; F). \end{aligned}$$

Let us prove $(a)_s^j$ for $s \in \mathbf{R}$ and $j < k$. We can assume that $\overline{U_t - U_s}$ is compact for any $s \leq t$ by replacing X by $\text{supp } F$. Consider the distinguished triangle

$$\mathbf{R}\Gamma_{X - U_t}(F) \rightarrow F \rightarrow \mathbf{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

^{*6}This condition might be deduced from (iii).

Applying to this triangle the functor $(\cdot)_{\overline{U}_t}$, we have

$$(\mathrm{R}\Gamma_{X \dashv U_t}(F))_{\overline{U}_t} \rightarrow F_{\overline{U}_t} \rightarrow \mathrm{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

Again applying to this triangle the functor $\mathrm{R}\Gamma(X; \cdot)$, we have

$$(6.1) \quad \mathrm{R}\Gamma(X; (\mathrm{R}\Gamma_{X \dashv U_t}(F))_{\overline{U}_t}) \rightarrow \mathrm{R}\Gamma(\overline{U}_t; F) \rightarrow \mathrm{R}\Gamma(U_t; F) \xrightarrow{+1}.$$

By hypotheses (iii)–(v), it follows that

$$\left(H_{X \dashv U_t}^j(F) \right)_{\partial U_t} = 0$$

for any $j \leq k$ and then

$$H^j(X; (\mathrm{R}\Gamma_{X \dashv U_t}(F))_{\overline{U}_t}) = 0 \quad \text{for any } j \leq k.$$

^{*7}Thus for any t , we have

$$(6.2) \quad H^j(\overline{U}_t; F) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for any } j < k,$$

$$(6.3) \quad H^k(\overline{U}_t; F) \hookrightarrow H^k(U_t; F).$$

Therefore, we have by (iv) and (6.2),

$$\begin{aligned} \varinjlim_{t>s} H^j(U_t; F) &\cong H^j\left(\bigcap_{t>s} U_t; F\right) \\ &= H^j(\overline{U}_s; F) \\ &\xrightarrow{\sim} H^j(U_s; F) \end{aligned}$$

for any $j < k$, and $(a)_s^j$ follows. Similarly, we have by (iv) and (6.3),

$$\begin{aligned} \varinjlim_{t>s} H^k(U_t; F) &\cong H^k\left(\bigcap_{t>s} U_t; F\right) \\ &= H^k(\overline{U}_s; F) \\ &\hookrightarrow H^k(U_s; F) \end{aligned}$$

and $(a')_s^k$ follows. □

Remark 6.2. Lemma 5.2 follows from Proposition 6.1. Indeed, putting $X =]-\infty, b[$, $F = G|_{I_b}$, and $U_t = I_t$, we find that

(i) $\bigcup_{s<t} I_s = \bigcup_{s<t}]-\infty, s[=]-\infty, t[= I_t$ for all $t \in \mathbf{R}$.

(ii) If $s \leq t$, then $\overline{I_t - I_s} \cap \mathrm{supp} G = \overline{[s, t[} \cap \mathrm{supp} G = [s, t] \cap \mathrm{supp} G$ is compact.

(iii) For $s \in \mathbf{R}$, we have

$$Z_s = \bigcap_{t>s} \overline{I_t - I_s} = \bigcap_{t>s} \overline{[s, t[} = \bigcap_{t>s} [s, t] = \{s\},$$

and

$$Z_s - I_t = \{s\} -]-\infty, t[= \begin{cases} \{s\} & \text{if } s = t, \\ \emptyset & \text{if } s < t. \end{cases}$$

^{*7} この辺の議論はきちんと結めないといけない。

By hypothesis of Lemma 5.2, for all $s \in \{s\}$,

$$H_{X \leftarrow I_s}^j(F)_s = H_{[s, \infty[}^j(F)_s = 0.$$

Then we can apply Proposition 6.1, and we have the isomorphisms for all $t \in [a, b[$ and $j < k$,

$$H^j(I_b; F) = H^j\left(\bigcup_s I_s; F\right) \xrightarrow{\sim} H^j(I_t; F)$$

and the injections

$$H^k(I_b; F) = H^k\left(\bigcup_s I_s; F\right) \hookrightarrow H^k(I_t; F)$$

7 Relative Microlocal Morse Theory via Truncated Microsupport

Let $f: Y \rightarrow X$ be a morphism of manifolds, and φ a real valued function of class C^1 . For a real number $t \in \mathbf{R}$, we set

$$\begin{aligned} Y_t &:= \{y \in Y ; \varphi(y) < t\} = \varphi^{-1}(]-\infty, t[), \\ \tilde{Y}_t &:= \{y \in Y ; \varphi(y) \leq t\} = \varphi^{-1}(]-\infty, t]). \end{aligned}$$

Let us define the following morphisms:

$$\begin{array}{ccc} Y_t & \xrightarrow{j_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X, \end{array} \quad \begin{array}{ccc} \tilde{Y}_t & \xrightarrow{\tilde{j}_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X. \end{array}$$

Proposition 7.1. Let $G \in D^b(Y)$, and assume $f_t: \text{supp}(G) \cap \overline{Y}_t \rightarrow X$ is proper for all $t \in \mathbf{R}$. Let k be an integer and t_o a real number.

(i) Assume that, for any $y \in Y - \tilde{Y}_{t_o}$, we have

$$d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^*X \right) \right).$$

Then

$$\begin{aligned} \text{(a)} \quad & R^j f_* G \cong R^j f_{t*} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o, \\ \text{(a}_k\text{)} \quad & R^k f_* G \hookrightarrow R^k f_{t*} j_t^{-1} G \quad \text{for } t > t_o, \\ \text{(a')} \quad & R^j \tilde{f}_* G \cong R^j \tilde{f}_{t*} \tilde{j}_t^{-1} G \quad \text{for } j < k \text{ and } t \geq t_o, \\ \text{(a}'_k\text{)} \quad & R^k \tilde{f}_* G \hookrightarrow R^k \tilde{f}_{t*} \tilde{j}_t^{-1} G \quad \text{for } t \geq t_o, \\ \text{(b)} \quad & \text{SS}_k(Rf_* G) \subset f_\pi \left(f_d^{-1} \left(\text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ & = f_\pi f_d^{-1} (\text{SS}_k(G)). \end{aligned}$$

(ii) Assume that, for any $y \in Y - \tilde{Y}_{t_o}$, we have

$$-d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^*X \right) \right).$$

Then

$$\begin{aligned}
(c) \quad & R^j f_! G \cong R^j f_{t!} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o, \\
(c_k) \quad & R^k f_! G \hookrightarrow R^k f_{t!} j_t^{-1} G \quad \text{for } t > t_o, \\
(c') \quad & R^j \tilde{f}_! G \cong R^j \tilde{f}_{t*} \tilde{j}_t^! G \quad \text{for } j < k \text{ and } t \geq t_o, \\
(c'_k) \quad & R^k \tilde{f}_! G \hookrightarrow R^k \tilde{f}_{t*} \tilde{j}_t^! G \quad \text{for } t \geq t_o, \\
(d) \quad & \text{SS}_k(Rf_! G) \subset f_\pi \left(f_d^{-1} \left(\text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\
& = f_\pi f_d^{-1} (\text{SS}_k(G)).
\end{aligned}$$

Proof. Let $\tilde{f}$ be the morphism $(f, \varphi): Y \rightarrow X \times \mathbf{R}$, and let q and $\tilde{\varphi}$ be the first and second projections defined on $X \times \mathbf{R}$ respectively. Then $\tilde{f}$ is proper on $\text{supp}(G)$ and $f = q \circ \tilde{f}$.^{*8} Let i_t be the open embedding $X \times]-\infty, t[\hookrightarrow X \times \mathbf{R}$ and $\tilde{i}_t$ the closed embedding $X \times]-\infty, t] \hookrightarrow X \times \mathbf{R}$. Set $q_t = q \circ i_t$ and $\tilde{q}_t = q \circ \tilde{i}_t$. These morphisms can be visualized by the following diagrams.

$$\begin{array}{ccc}
Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} & \xrightarrow{q} & X, \\
j_t \uparrow & & \uparrow i_t & \nearrow q_t & \\
Y_t & \xrightarrow{\tilde{f}_t} & X \times]-\infty, t[& &
\end{array}
\quad
\begin{array}{ccc}
Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} & \xrightarrow{q} & X. \\
\tilde{j}_t \uparrow & & \uparrow \tilde{i}_t & \nearrow \tilde{q}_t & \\
Y_t & \xrightarrow{\tilde{f}_t} & X \times]-\infty, t] & &
\end{array}$$

Then, setting $\tilde{G} := Rf_* G$, we have

$$\begin{aligned}
Rf_* G &\cong Rq_* \tilde{G}, & R\tilde{f}_* G &\cong R\tilde{q}_* \tilde{G}, \\
Rf_{t*} j_t^{-1} G &\cong Rq_{t*} i_t^{-1} \tilde{G}, & R\tilde{f}_{t*} \tilde{j}_t^{-1} G &\cong R\tilde{q}_{t*} \tilde{i}_t^{-1} \tilde{G}, \\
Rf_{t!} j_t^{-1} G &\cong Rq_{t!} i_t^{-1} \tilde{G}, & R\tilde{f}_{t*} \tilde{j}_t^! G &\cong R\tilde{q}_{t*} \tilde{i}_t^! \tilde{G}.
\end{aligned}$$

Thus taking j -th cohomology, we have

$$\begin{aligned}
R^j f_* G &\cong R^j q_* \tilde{G}, & R^j \tilde{f}_* G &\cong R^j \tilde{q}_* \tilde{G}, \\
R^j f_{t*} j_t^{-1} G &\cong R^j q_{t*} i_t^{-1} \tilde{G}, & R^j \tilde{f}_{t*} \tilde{j}_t^{-1} G &\cong R^j \tilde{q}_{t*} \tilde{i}_t^{-1} \tilde{G}, \\
R^j f_{t!} j_t^{-1} G &\cong R^j q_{t!} i_t^{-1} \tilde{G}, & R^j \tilde{f}_{t*} \tilde{j}_t^! G &\cong R^j \tilde{q}_{t*} \tilde{i}_t^! \tilde{G}.
\end{aligned}$$

Therefore, it is enough to prove the result for $\tilde{G}$, q , i_t , $\tilde{\varphi}$ instead of G , f , j_t , φ .

(i) It follows from the hypothesis that

$$(7.1) \quad \text{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\}) \subset T^*X \times \{(t; \tau); \tau \leq 0\}.$$

Indeed, by hypothesis, for any $(y; \eta) \in \text{SS}_k(G)$ and $(f(x); \xi) \in Y \times_X T^*X$, we have

$$(7.2) \quad d\varphi(y) \neq \eta + f^* \xi.$$

If $(x, t; \xi, \tau)$ is a point of $\text{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\})$, then, by the formula (3.2), there exists $(y; \eta) \in \text{SS}_k(G)$ such that

$$\tilde{f}(y) = (x, t) \text{ and } \eta = \tilde{f}^*(\xi, \tau).$$

^{*8}This can be proved as follows. If K is a compact set of $X \times \mathbf{R}$, $\tilde{f}^{-1}(K)$ is compact.

For this η , we have

$$\eta = \tilde{f}^*(\xi, \tau) = f^*\xi + \varphi^*\tau = f^*\xi + \tau d\varphi(y).$$

If $\tau > 0$, we get

$$d\varphi(y) = \frac{1}{\tau}\eta - \frac{1}{\tau}f^*\xi$$

and this contradicts (7.2) because $\text{SS}_k(G)$ is conic.

By (7.1), Let U be an open subset of X contained in a local chart, and convex in such a chart. By Proposition 6.1, we have isomorphisms for all $t > t_o$ and $j < k$

$$(7.3) \quad H^j(U \times \mathbf{R}; \tilde{G}) \xrightarrow{\sim} H^j(U \times]-\infty, t[; \tilde{G})$$

and have an injection for all $t > t_o$

$$(7.4) \quad H^k(U \times \mathbf{R}; \tilde{G}) \hookrightarrow H^k(U \times]-\infty, t[; \tilde{G}).$$

This proves the result. □

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